

Deeds: One-Stop Solution for All Your Digital Electronics Design Needs

Deeds, short for 'Digital Electronics Education and Design Suite,' is a set of educational tools for digital Electronics that adopt learn-by-doing approach. Developed by the University of Genoa, Italy, this design suite comes bundled in this month's Electronics for You Plus DVD. We hope you find it as useful as we did

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Deeds was conceived by the University of Genoa, Italy, as a suite of simulators, tools and learning material for digital electronics design. It helps students acquire theoretical foundations, analysis and problem solving capabilities all over the design electronics domain, and practical synthesis and design skills. Its approach is characterised by the mantra of "learning-by-doing."

Deeds covers the following areas of digital electronics:

1. Combinational logic networks ranging from simple gates to decoders, encoders, multiplexers and demultiplexers
2. Sequential logic networks from a simple flip-flop to the slightly less simple registers and counters
3. Finite-state machine design and microcomputer programming (at Assembly level) along with microcomputer interfacing
4. Field-programmable gate array programming

Tools and features

Simulators are integrated around two HTML browsers: an HTML main browser to navigate on the Internet, where students can find lessons, exercises and laboratory assignments, and an assistant browser, which assists

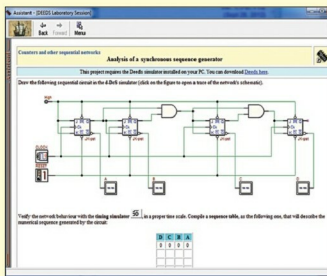


Fig. 1: Assistant browser is opened, showing a page with a problem assignment

students in their work. These browsers allow you to launch all the tools and also interact with them. Hence the browsers can now control editors and simulators, and realise a true interaction between theory and practical results. The main Web browser of Deeds, when activated, shows an HTML page that allows you to connect to the Deeds website and to the 'online' learning material developed at DIBE (University of Genoa).

The main browser has been developed around the standard Microsoft WebBrowser component extended to support all the functions required

by the Deeds environment. It is mainly used to connect to the sites containing the learning material. The browser supports all the features that the user can expect to find, including JAVA Virtual Machine, VBScript, JavaScript and XML support.

The assistant HTML browser has characteristics similar to those of the main browser

but it is customised to assist students in their work. This browser is used to open lessons, exercises and laboratory assignments. Similar to the main browser, the assistant browser has been conceived around the standard Microsoft WebBrowser component. All the objects visualised on a webpage can be made 'active.'

There is a schematic digital circuit editor in addition to a component data-sheet support. The software houses an interactive circuit animator (to experiment with components and simple networks directly on the schematics), an interactive logic simulator (with a tim-